00. Class Introduction

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▼ Class Information

▼ Time and Place

- Wednesdays, 9:10 AM–12:00 PM
- EE-II 104, National Taiwan University (NTU)

▼ Notion Page: ✌🏾 114-2 SoCV

> Starting this year, we will migrate our lecture notes from Pages/PDF to Notion.
> The 113-2 lecture notes (PDF) will also be provided as reference material.

▼ NTU COOL

- Used primarily for announcements and discussions.
- NOT open for auditing students

▼ Contact Email (for all purposes): ntueesocv@googlegroups.com

- Please use this email address for all course-related communication.

▼ My Office (EE-II 444)

- Email: cyhuang@ntu.edu.tw (responses may be slow)
- Other online ID: ric2k1
  - Please identify yourself. Otherwise, I may not respond.
- Office hours: by appointment (I am not in the office most of the time)

▼ TAs

- 黃思維: EE-II 353, Mon 1:00–3:00 PM
- 黃博聞: EE-II 353, Tue 2:00–3:00 PM

⇒ To schedule TA hours, send an email to ntueesocv with clear title and description.

▼ Objectives of this Class

▼ Understand core formal verification techniques

- Formal modeling and property specification
- Binary Decision Diagrams (BDDs)
- Boolean satisfiability (SAT) solving
- Model checking and reachability analysis

▼ Appreciate practical applications of formal methods

- Formal-assisted logic synthesis
- Satisfiability vs. optimization problems

▼ Develop system-level intuition

- Why is verification both essential and difficult?
- How do tools and algorithms help in practice?

▼ Establish a healthy and effective learning mindset

- With rapid advances in AI, what/how should I learn, and why?
- Be self-motivated, self-driven, and responsible for learning

😨 **Although we will mention SystemVerilog (SV) and the Universal Verification Methodology (UVM), we will not cover them in detail in this course.**

- Some historical reasons...
- It is hard to assess learning outcomes for these practical skills.

▼ Key Innovation (Experiment) this Year

😲 **Everyone should maintain a "My Learning Notes (我的學習筆記)"**

▼ What you will do

- During and after each lecture, the instructor (Ric) will raise research questions based on the teaching material.
- You will investigate these questions and write up your findings in your notes. Keep one note for the entire semester.
- There is no strict format, but we will provide a template for reference.

▼ Format requirements

- There is no restriction on the tool or file format, but your notes **must be accessible on the internet**.
- Acceptable examples include, but not limited to, Notion, NotebookLM, Heptabase, Google Docs/Slides, or a web service you build.
- Your notes must have **one single URL** that we can open.
- We will collect this URL by the **3rd meeting**.

▼ AI tools are encouraged

- You are encouraged to use AI tools in any part of your work, including drafting content, prototyping ideas/algorithms, writing code, and analyzing experimental data.

- To ensure everyone has baseline access, each registered student will receive **2 months of Cursor Pro credits**.
- If you remain enrolled after the midterm (i.e., not dropping the course), you will receive an additional **3 months** of credits.

💪 ▼ **AI is just a vehicle. You are the driver.**

- Don't let AI auto-pilot where you go. You should direct it (with proper prompts) to explore the right information you need.

▼ Grading cadence

- Notes should be updated weekly.
- TAs will grade the notes **every two weeks**.
- The first grading will be in **Week 3**, covering Weeks 1–2.
  - For Week 1, all research questions must be answered.
  - For Week 2, questions may be in progress.
- There will be **8** grading rounds in total, including the final round in **Week 17**.
- Detailed criteria will be announced later and may be adjusted for fairness.

😊 ▼ **Learning is your responsibility, not ours.**

- Put in more effort, learn more, and good grades will follow.
- We are here to support and facilitate learning, not to supervise or monitor.

▼ Class Schedule (TBC)

Wk	Date	Topic
1	02/25	00. Class Introduction 01. Introduction to Design Verification

2	03/04	01. Introduction to Design Verification 02. Formal Modeling and Property Specification
3	03/11	03. Intro to Binary Decision Diagram
4	03/18	04. Advanced BDD Techniques
5	03/25	05. BDD-Based Verification
6	04/01	05. BDD-Based Verification
7	04/08	06. Introduction to Boolean Satisfiability
8	04/15	07. Advanced SAT Techniques
9	04/22	08. Bounded Model Checking; 09. Unbounded Model Checking
10	04/29	10. Property Directed Reachability
11	05/06	10. Property Directed Reachability
12	05/13	11. Applications of SAT on Logic Synthesis
13	05/20	11. Applications of SAT on Logic Synthesis
14	05/27	12. From SAT to Optimization
15	06/03	13. Special Topics (TBD)
16	06/10	Final Exam Week (No Exam)
17	06/17	Final Project Presentation / Demo

▼ Class Overview

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▼ Grading

▼ My Learning Notes — 60%

- There is no single “correct” answer for the learning notes.
- Grading will be based on our professional judgment, with an emphasis on **effort** and **research depth**.
- If you are not comfortable with this type of qualitative evaluation, please reconsider enrolling in this class.

- Detailed grading criteria will be announced later and may be adjusted for fairness.

▼ Final project — 40%

- Research problems will be introduced during the class. You may choose one of them or propose your own topic.
- Projects may be individual or team-based, depending on the scope.
- Milestones
  - Project proposal and team formation: **end of April (TBD)**
- Requirements
  - (SW) Implementation and experiments are required.
  - Pure literature surveys (paper-review-only projects) are not allowed.
- Example topics
  - Formal modeling and abstraction for gate-level or system-level design verification
  - Equivalence checking or model checking problems
  - Formal-assisted ECO or logic optimization
  - CAD Contest Problem #A (Verification/Synthesis problem by Cadence)

👉🏿 **No midterm or final exams**

Final grades will be linearly adjusted.

The instructor will determine the class average and standard deviation.

🤔 ▼ Before you decide to take this course...

▼ Make sure you can keep up with “weekly” learning notes

- Each bi-weekly grading round is worth **7.5 points** of your term grade. Missing one will hurt.

▼ Make sure you accept that learning is your responsibility

- It is easy to generate “answers” by pasting the research questions into AI tools.
- But if you do not spend time thinking, extending the questions, and digesting the material, you will not truly learn.
- Please do not turn your notes into something you do not understand. That’s a waste of life.

▼ If you decide to sit in this class...

- Welcome. These lecture notes (notion page) are publicly available online.
- However, we do not provide NTU COOL access for sit-in students, since NTU COOL is used for official announcements and course discussions.

▼ About video recording

- I will try to record the screen for each class meeting and publish the video on YouTube.
- Occasionally, the recording may fail, or the audio/video quality may be suboptimal.
- Videos are provided **as is**, with no additional editing or post-processing.